Affective Valence and Temporal Framing: A Unified Generative Model of the Temporal Structure of Emotion

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Abstract

Emotions are temporally structured: regret concerns what has happened, anxiety what will happen, joy what is happening now. Three strands of the active-inference literature each formalise one temporal face of affect: backward valence as the negative rate of change of variational free energy (Joffily and Coricelli, 2013), present valence as reward prediction error (Pattisapu et al., 2025), and forward valence as the affective charge of policy revision (Hesp et al., 2021). Yet each alone is *insufficient*: none spans the full temporal range, unifies these quantities into a taxonomy of emotion, or models how an agent actively selects the temporal horizon that dominates its own affect. We develop a single factored generative model in which a hidden *temporal-frame* state (past/present/future) redistributes precision across horizons, so that framing emerges from expected-free-energy minimisation. We motivate the architecture from the psychology of mental time travel, the graded, cost-bounded regulation of how deeply memories and plans are unrolled, and show that a single *depth regulator* with two temporal faces reproduces the classical split between rumination (past-directed) and worry (future-directed). Affect is a *readout layer* over a task-specific generative model, and the three channels are general operators on its inference dynamics. Empirically, driven through real experience-sampling sequences from two independent samples, the full model predicts next-moment affect out of sample better than linear autoregressive baselines, roughly doubling one-step skill where affect carries volatile dynamics (held-out $R^2 = 0.19$ vs 0.09), the out-of-sample lead holding on a second, independent sample, and an inertia ablation shows the gain comes from the model's own forward temporal dynamics rather than a persistence prior. On a static gamble it recovers the standard happiness equation ($n = 14{,}803$ held out) where each predecessor channel alone falls short, reaching that ceiling without injecting expected value as a regressor (the forward/expected-free-energy channel supplies it by construction, a +30% gain over a reward-prediction-error channel). The same integration then supports capabilities the single-channel accounts lack: a latent temporal-frame state that tracks an independently measured symptom it is never shown ($r = 0.17$); counterfactual emotions with a behavioural signature a reward-only model cannot produce (regret drives choice-switching over and above one's own outcome, $t = 10.4$, replicated on a second dataset); and, as a demonstration awaiting longitudinal test, a mood layer that produces a diathesis-stress route to depression from the conjunction of vulnerability and chronic stress. Throughout we separate predictive claims (the channel integration) from generative ones (counterfactual and hedonic-asymmetry mechanisms).

Contents

1 Introduction

Humans have a distinctive capacity for mental time travel. They simulate past, present, and future events to guide present decisions (Suddendorf and Corballis, 1997; Schacter and Addis, 2007). Moment-to-moment shifts of temporal orientation play out against more stable temporal dispositions, letting people balance long horizons against the demands of the immediate. These shifts are inseparable from emotion: regret is directed at the past, hope at the future, surprise refocuses on the present. A central question is how past, present, and future are *instrumentally recruited* to guide behaviour toward preferred outcomes, and how emotions participate in that recruitment.

We approach this within predictive processing, on the view that emotional experience is an integral component of the inferential processes organising perception, learning, and action (Barrett, 2017; Seth and Friston, 2016). On this view affect is grounded in the agent's success at minimising prediction error. It can be past-oriented, as the derivative of the trajectory of model fit (Joffily and Coricelli, 2013). Improving fit yields positive valence, deteriorating fit negative. It is also forward-looking. Expected free energy (EFE), minimised in policy selection (Parr et al., 2022), evaluates how well future observations will match preferences, and the "affective charge" of policy revision is itself a valence signal (Hesp et al., 2021). A present-oriented reward-prediction-error channel completes the picture (Pattisapu et al., 2025).

These accounts establish that affect has temporal structure, but they are *insufficient* in three respects. First, none unifies the variational/expected free-energy (VFE/EFE) distinction into a temporal taxonomy of emotion. Second, none models the single-agent mechanism by which an individual selects which temporal horizon of free energy dominates its affect, the mechanism Hinrichs et al. (2025) raised for dyads under the banner of "temporal aiming", but left open within the agent. Third, each formalises only one temporal face of affect in isolation.

We close all three gaps. We (i) motivate the architecture from the psychology of mental time travel and cost-bounded simulation (§2); (ii) propose the VFE/EFE mapping as a unified theory of affective temporal direction and derive a three-channel valence decomposition (§3, §5); (iii) formalise counterfactual temporal framing as precision redistribution across horizons within a single factored generative model (§4); and (iv) show empirically that this integration is not merely conceptual. The integrated readout recovers prior single-channel models as special cases and improves on them where temporal structure exists (§9). Throughout, we separate claims that are *predictive* (the channel integration) from those that are *generative* (the counterfactual and hedonic-asymmetry mechanisms).

2 Mental time travel and the depth regulator

For much of everyday cognition, agents mentalise goals and the paths toward them, detaching from the present to simulate the future. Because remembering and future thinking recruit a shared constructive machinery (Schacter and Addis, 2007), these simulations are built out of memory. They must be precise enough to guide action, yet they are built with finite resources, a constraint that becomes acute as we juggle timescales from the immediate to the lifelong. Planning over a longer horizon prima facie requires more state-to-state transitions than a shorter one, but transitions are not free. Tree search over policies grows with branching factor and depth, so exhaustive planning is intractable for any bounded system, and each rollout step introduces approximation error that compounds along the trajectory.

2.1 Scope, and the role of optimism

The empirical literature on future-oriented cognition shows how humans handle this. At long horizons, little fine-grained arbitration takes place. Agents rely on broad, optimistic beliefs that preserve orientation without detailed simulation. Rather than constructing every future at fixed resolution, they "zoom" in and out across temporal scales; the number and resolution of intermediary steps vary with environmental complexity, novelty, uncertainty, and affective load. Near events are represented concretely, distant events abstractly and schematically (Trope and Liberman, 2010). Optimism is computationally sensible here. The optimal path to a distant goal cannot be specified in advance, and filling in that far-off detail tends to provoke anxiety rather than yield mastery. On the broaden-and-build account (Fredrickson, 2001), positive affect widens the field of available thought and action and is less action-specific than negative affect, so future aiming is supported by beliefs that orient without specifying the future in full. Optimism here is the healthy, high-precision setting rather than a universal default: under low positive-belief precision the same forward machinery yields anxious rather than hopeful prospection, a contingency we make explicit in the model (§4) and test directly (§9.4).

2.2 Two logics of action and the identity pole

Decision theory distinguishes two families of action selection (March, 1994). On the *logic of appropriateness*, the agent asks what kind of situation this is and what a person like them does here, recognising a situation and completing a pattern from a self-representation, rather than computing outcomes. On the *logic of consequences*, it lays out candidate action–outcome sequences and scores them against goals. The two differ in how deeply the plan is unrolled. Appropriateness conditions choice on slow, temporally extended beliefs about who one is and stays shallow; consequence-based control descends toward episode-level simulation. This maps a long-standing continuum between habitual (model-free) and goal-directed (model-based) control, but recasts it in *temporal* terms. A past-oriented mode leans on regularities slowly distilled from experience, a future-oriented mode is less tied to the current context.

2.3 Retrieval constraints, interoception, and overgeneral memory

How far the past can be recruited to serve the present is set by how autobiographical memory is actually constructed, and this is what a past-directed control operator in the model must respect. In the Self-Memory System (Conway and Pleydell-Pearce, 2000), autobiographical retrieval is a negotiation between *correspondence* (fidelity to experience-near sensory records) and *self-coherence* (pressure to keep memories consistent with the conceptual self); the working self gates construction as retrieval descends the autobiographical hierarchy. Whether the agent can reach a specific, self-congruent episode depends on its current affective and interoceptive state and its emotional granularity. Coarse, negatively valenced states narrow access to specific, positively toned material. The failure mode of the shallow setting is *overgeneral memory* (Williams et al., 2007). When coherence wins over correspondence, the cue is answered from the kind of person one takes oneself to be, not from an episode. This fixes what the past-directed operator of §4 must do. There the RECALL action pulls represented valence toward an identity-anchored target whose sign is set by the agent's positive-belief precision (π_{pos}, the confidence of the slow mood belief that outcomes will be favourable; formalised in §4), so a coarse, negatively biased self answers a cue with overgeneral, negatively toned material, the failure mode just described. The abstract negotiation becomes one control variable in the model, how deeply retrieval descends the hierarchy.

2.4 One depth regulator, two temporal faces

These constraints force a control problem. Simulating the past or the future is unbounded and metabolically costly, so something must set how far any given memory or plan is unrolled. Left unregulated the system fails in one of two directions: it over-simulates, which is intractable and, as above, anxiogenic, or it under-simulates, which yields overgeneral memory looking back and absent foresight looking forward. We call the quantity that sets this depth the *depth regulator*, and it is what the rest of the model controls. It is one regulator rather than two because the same knob, how far to descend into constructed episodes, operates in both temporal directions. It weighs identity fit against computational cost, and it has two temporal faces. On the past side, the agent arbitrates between self-coherence and correspondence; its healthy shallow setting is a positively biased working self. On the future side, the healthy shallow setting is orientation without detailed simulation; a plan is committed to only once computed and broken into steps, with depth bounded by cost. The two directions of failure recover the classical split (Nolen-Hoeksema et al., 2008): *rumination* (past-directed, self-focused) when a habitual, identity-congruent bias overrides the escape that consequence-based control would favour; *worry* (future-directed, anticipatory) when a bias toward reward overrides the cost that limits forward simulation (Ehring and Watkins, 2008). This double bind is the computational core of the chronic-stress pathology of §7. The remainder of the paper gives the depth regulator a formal treatment in the language of free energy. In the model that follows, its continuous depth is discretized into a three-valued temporal-frame factor and six framing actions that redistribute precision over it. The discretization is a simplification, flagged again in the limitations, and does not assert that regulated depth is intrinsically discrete.

3 The temporal structure of free energy

3.1 Active inference and attention as precision

Active inference holds that agents encode a generative model relating hidden states to observations and minimise free energy across perception, learning, and action. Prediction errors, discrepancies between expected and sensed observations, drive belief updates, weighted by *precision*, the gain that determines which errors shape belief. Precision-weighting is attention (Feldman and Friston, 2010), and agential control over it requires a metacognitive architecture in which higher-order states modulate which lower-level processes receive precision (Sandved-Smith et al., 2021). It is this second-order control of precision over temporal horizons that we formalise as temporal framing.

3.2 VFE as backward-looking evaluation

For a generative model $p(o, s)$ with approximate posterior $q(s)$,

$$F = \underbrace{-\mathbb{E}_{q(s)}[\ln p(o \mid s)]}_{\text{accuracy}} + \underbrace{D_{\mathrm{KL}}[q(s) \,\|\, p(s)]}_{\text{complexity}}, \tag{1}$$

which scores how well the *current* model fits *past* observations. Joffily and Coricelli (2013) showed that its temporal derivative is a valence signal: decreasing F (improving fit) is positive valence, increasing F negative. The second derivative differentiates its dynamics: accelerating improvement, decelerating improvement toward a plateau, and their negatives give a family of affective states from a single backward-looking quantity. These are dynamics of *past* model fit and are distinct from the forward-looking hope and fear of §5, which evaluate *future* observations, even where ordinary language reuses the words.

3.3 EFE as forward-looking evaluation

For a policy π producing predicted observations,

$$G(\pi) = \sum_m \Big[\underbrace{D_{\mathrm{KL}}[q(o_m \mid \pi) \,\|\, p(o_m \mid C_m)]}_{\text{risk}_m} + \underbrace{\mathbb{E}_{q(s)}[H[p(o_m \mid s)]]}_{\text{ambiguity}_m} \Big], \tag{2}$$

where C_m encodes preferences. EFE asks not "did I predict well?" but "will I predict well?". Its risk term is expected value relative to preference; this is the quantity a forward affective channel reads out, and the quantity that supplies *anticipated value* when the channels are placed on a decision task-model.

3.4 Constructionism and metacognition

By *agent* we mean the organism modelled as an active-inference system, not a homunculus inside it, and we take no stance on phenomenal consciousness; the distinction we rely on is between levels of control, not between conscious and unconscious experience. Running automatically, without second-order intervention, the base perception–action loop reproduces a flow-like present engagement with the world (Rietveld and Kiverstein, 2014). Made multiscale, it affords a constructionist account of emotion (Barrett, 2017). Emotions are concepts moulded onto core processes of allostatic regulation and interoceptive monitoring (Seth and Friston, 2016; Stephan et al., 2016). Metacognitive control of attention, a covert mental action that modulates which processes receive precision, requires a second-order layer (Sandved-Smith et al., 2021), higher-order states setting lower-level precision, and it is here that counterfactual temporal framing lives. Remembering and imagining are a single offline generative operation, the same model run over past or future states instead of present ones (Schacter and Addis, 2007).

4 The generative model

We adopt a factored POMDP with hidden state $s = (v, e, f)$: **valence** $v \in \{0, \ldots, K-1\}$ at granularity K; **energy** $e \in \{0, \ldots, M-1\}$, an interoceptive resource estimate; and **temporal frame** $f \in \{\text{PAST}, \text{PRESENT}, \text{FUTURE}\}$, the active counterfactual orientation. Three observation modalities, external feedback o_{ext}, interoception o_{int}, and felt valence o_{val}, couple to these factors. The frame f is a hidden state the agent *infers*, not a label it assigns. The posterior $q(f \mid o)$, marginalised over v and e, is a direct readout of believed temporal orientation.

Two quantities in this paper carry the name valence, and separating them removes an apparent circularity. The hidden state v is *represented* valence, the agent's coarse posterior belief about its own affective condition, inferred jointly from external feedback, interoception, and a felt-valence signal, and used to gate retrieval and to report. The three *valence channels* of §3 are not states but signals read off the dynamics of inference itself: the backward channel is the rate of change of variational free energy ($-dF/dt$), the present channel a reward prediction error, the forward channel the affective charge of policy revision. These channels generate the felt-valence observation o_{val}, and the state v is what the agent infers from it. Valence is therefore neither simply given nor only a derivative of free energy. The dynamic signals drive the felt observation, and v is the belief the agent forms about its affective state from that observation together with its other evidence. This split is what we mean by affect as a readout layer over a task-specific generative model. Figure 1 shows the factor graph. To read it, note that the agent receives three observations (the coloured outcome squares: external feedback, interoception, and felt valence), infers the three hidden factors that generate them through a likelihood $\mathbf{A}$ (valence, energy, temporal frame), and moves each factor through a controllable transition $\mathbf{B}$ by selecting a policy, scored by expected free energy G. Red arrows are the message passing that carries beliefs between them. It is one task-specific model, and the three affective channels are three readings taken from this single loop rather than three separate models.

4.1 Framing actions as temporal-precision operators

Each action defines a transition $\mathbf{B}_a = \mathbf{B}_v^{(a)} \otimes \mathbf{B}_e^{(a)} \otimes \mathbf{B}_f^{(a)}$, whose frame block $\mathbf{B}_f^{(a)}$ is a *temporal-precision operator* that redistributes probability across horizons. Six actions realise the depth regulator:

RECALL assigns 70% self-transition to PAST; $\mathbf{B}_v^{\text{RECALL}}$ pulls valence toward a *bidirectional* target $v_{\text{recall}} = 0.2 + 0.6\alpha$ with $\alpha = \sigma(\pi_{\text{pos}} - \theta)$, so high positive-belief precision π_{pos} recalls a positive past (narrative stabilisation) and low π_{pos} a negative one (rumination). It is the computational realisation of *shallow*, identity-anchored retrieval (§2.4).

ENGAGE assigns 75% to PRESENT, maintaining precision on current sensory input through perception–action coupling.

FUTURATE assigns 90% to FUTURE (narrative inertia); $\mathbf{B}_v^{\text{FUTURATE}}$ pulls valence toward a *precision-gated* target $v_{\text{fut}} = 0.2 + 0.6\alpha$, symmetric with RECALL, so high positive-belief precision yields hopeful prospection and low precision yields anxious, worry-type prospection rather than fixed optimism. Its energy cost brakes it into short reactive planning bursts.

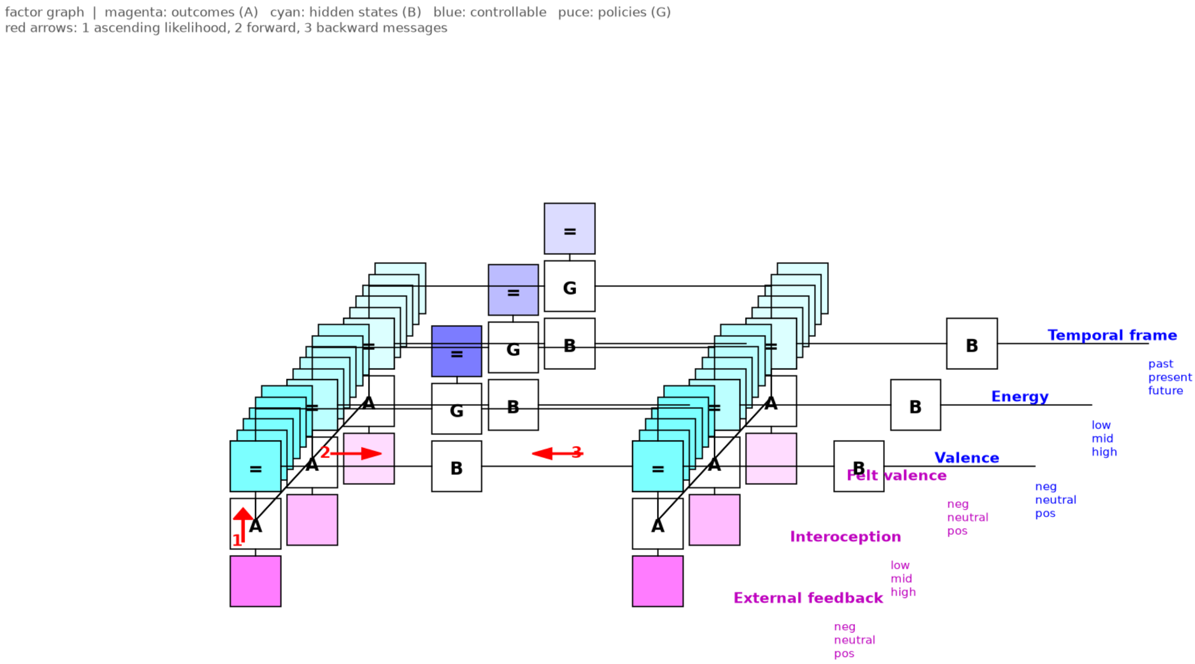

Figure 1: **Factor graph of the generative model** (SPM convention: magenta = outcome modalities with likelihood $\mathbf{A}$; cyan = hidden-state factors with transition $\mathbf{B}$; controllable factors are stacks with policy and expected-free-energy G nodes; red arrows = message passing). Three controllable factors (valence, energy, temporal frame) and three modalities integrate the backward, present, and forward affective channels in one model.

FEEL releases frame commitment (50–60% present) and models active interoceptive processing, reducing accumulated prediction error through body-state monitoring (Stephan et al., 2016).

DISSOCIATE distributes probability roughly uniformly across frames, modelling *temporal unmooring* (Vannikov-Lugassi and Soffer-Dudek, 2018).

ABSTRACT assigns $\approx$ 60–80% to FUTURE, an effortful, ungrounded forward route that bypasses interoception and episodic recall.

$\mathbf{B}_{\text{frame}}$ matrices are additionally learned online via Dirichlet updates, so agents acquire personalised temporal-attention operators. Here σ is the logistic function and $\theta = 2$ the precision threshold; the exact frame- and valence-transition entries, the channel temperatures $\tau_{\text{model}}, \tau_{\text{reward}}, \tau_{\text{action}}$, and the mood-layer constants are given in the supplementary implementation (`generative_model.py`, `agent.py`).

4.2 Extensions and the mood layer

Two extensions capture affective pathology. *Asymmetric hedonic sensitivity* replaces the scalar reward sensitivity with a pair $(c_{\text{pos}}, c_{\text{neg}})$ scaling positive and negative preference entries separately, capturing dissociable reward and punishment processing (Eshel and Roiser, 2010; Pizzagalli, 2014). *Habit priors* add an additive log-prior $\mathbf{E}$ over policies (Da Costa et al., 2020), so $\ln \pi(a) = -\gamma G(a) + E_a$, modelling EFE-suboptimal habitual tendencies such as ruminative fixation (Watkins, 2008). A hierarchical *mood layer* (M5) infers π_{pos} over slow timescales as a categorical POMDP driven by accumulated believed-valence relative to neutral (Eldar et al., 2016), separating within-trial emotion from cross-trial mood (Hesp et al., 2021) and producing a self-reinforcing low-mood attractor under vulnerability and stress from Bayesian inference alone (§7).

Table 1 maps each choice to its source; full detail is in the supplementary design-justification document.

Table 1: **Design justification (excerpt).** Predictive (P) and generative (G) status flagged where the distinction applies.

Choice	Rationale	Source
Attention = precision (P)	framing is precision redistribution over horizons	Feldman and Friston (2010)
Backward channel (P)	$-dF/dt$ valence	Joffily and Coricelli (2013)
Present channel (P)	reward prediction error	Pattisapu et al. (2025)
Forward channel (P)	affective charge of policy revision (EFE)	Hesp et al. (2021)
Temporal-frame factor	mental time travel; shared memory/imagination machinery	Schacter and Addis (2007); Suddendorf and Corballis (1997)
Energy/interoception	allostatic body-budget	Seth and Friston (2016); Stephan et al. (2016)
RECALL (past)	self-memory system; overgeneral memory on failure	Williams et al. (2007)
Habit priors (E-vector)	habitual, EFE-suboptimal tendencies	Da Costa et al. (2020); Watkins (2008)
Hedonic asymmetry (G)	reward/punishment sensitivity dissociate	Eshel and Roiser (2010); Pizzagalli (2014)
Counterfactual emotions (G)	regret/relief from obtained vs foregone	Hesp et al. (2020)
Mood layer (M5)	mood/emotion timescale separation	Hesp et al. (2021); Eldar et al. (2016)
Effort/energy cost	effortful deliberation carries a metabolic cost	Treadway et al. (2009)

5 Three-channel valence and the temporal taxonomy

The distribution of attention across horizons is a second-order inference, and the three channels are read at successive stages of a single agent–environment cycle. Updating beliefs from a new observation exposes the backward channel, model valence. Comparing obtained against expected reward exposes the present channel, reward valence. Revising the policy exposes the forward channel, action valence.

5.1 The three channels

Model valence (backward) tracks $v_{\text{model}} = \tanh(-\Delta F/\tau_{\text{model}})$ (Joffily and Coricelli, 2013), modulating the model learning rate. *Reward valence* (present) is $v_{\text{reward}} = \tanh((U - \mathbb{E}[U])/\tau_{\text{reward}})$ over hedonic modalities (Pattisapu et al., 2025); interoceptive prediction errors drive allostatic regulation but are excluded from felt valence. *Action valence* (forward) is $v_{\text{action}} = \tanh(\text{AC}/\tau_{\text{action}})$ with $\text{AC} = (\boldsymbol{\pi}_{t-1} - \boldsymbol{\pi}_t) \cdot \mathbf{G}$ (Hesp et al., 2021), and feeds back on policy precision. This closes the emotion–action loop: policy revision changes expected free energy, action changes what is observed, and the resulting change in variational free energy feeds back through v_{action} onto the precision that governs the next policy.

5.2 Composite valence and channel disagreement

The channels combine as $V = \tanh(v_{\text{model}} + v_{\text{reward}} + v_{\text{action}})$. Agreement yields unambiguous satisfaction or distress; disagreement is informative. *Productive failure* ($v_{\text{model}} < 0$, $v_{\text{action}} > 0$): the outcome was bad but a better plan was found. *Unsettling success* ($v_{\text{reward}} > 0$, $v_{\text{action}} < 0$): good fortune that may not last. *Dawning anxiety* ($v_{\text{model}} \approx 0$, $v_{\text{reward}} \approx 0$, $v_{\text{action}} < 0$): nothing has happened, but plans are deteriorating.

5.3 A worked example

Take one trial of the gamble task-model of §9. The agent gambles and wins, and the win is larger than it expected. Read the channels in turn. The present channel compares obtained reward with its expectation; the surplus makes $v_{\text{reward}} > 0$. The backward channel registers that the win improved model fit, so F fell and $v_{\text{model}} > 0$. The forward channel registers that policy precision has shifted toward gambling and that this lowered expected free energy, so its affective charge $v_{\text{action}} > 0$. All three agree, the composite $V = \tanh(v_{\text{model}} + v_{\text{reward}} + v_{\text{action}})$ is strongly positive, and the state is unambiguous satisfaction. Now change one thing: let the gamble lose, but let the loss reveal that the safe option was the better policy. The present and backward channels turn negative (the outcome was bad and fit worsened) while the forward channel stays positive (a better plan was found). The channels disagree, and the composite reports the muted, forward-leaning affect labelled productive failure above. These three readings, taken at three stages of one inference cycle, are all the taxonomy requires.

5.4 The VFE/EFE temporal taxonomy of emotion

Emotions inherit their temporal direction from whichever quantity generates them, yielding a taxonomy we offer as an organising scheme rather than an essentialist classification; its partitioning within an agent depends on developmental and sociocultural history (Barrett, 2017). *Backward* ($-dF/dt$, the rate of change of VFE): regret, guilt, shame, pride, gratitude. *Present* (RPE): joy, anger, disgust, surprise. *Forward* (EFE): hope, fear, anxiety, anticipation. *Transitional* (VFE vs prior EFE): relief, disappointment, satisfaction. This grounds the temporal axis of appraisal theory (Smith and Ellsworth, 1985; Ortony et al., 1988) in a formal distinction rather than folk labels.

6 Prior models are special cases

Figure 2 shows the architecture as a stack of layers and where the prior models sit inside it. A task-specific generative model sits at the base: states, observations, policies, and expected free energy for whatever task the agent faces (the gamble task-model of §9, an experience-sampling affect model, any decision problem). On top of it sits the affective readout layer, the three channels, whose composite feeds a represented valence state and a slow mood layer, with the temporal-frame factor modulating precision across horizons. Each prior model is one layer of this stack, and our contribution is the integration, not any single channel.

The three predecessor accounts are *reduced sub-graphs* of our model (Figure 3, schematic): Joffily and Coricelli (2013) is a perception-only VFE model (one factor, no policy, backward channel only); Pattisapu et al. (2025) is a single-factor POMDP with policy selection (present channel, no temporal frame, no backward derivative); Hesp et al. (2021) is a deep hierarchical model carrying the forward affective-charge channel and a slow mood layer, but without a reward decomposition or a temporal-frame factor. Each predecessor contributes at most one of our affective channels, and we make the debt explicit: the mood layer that carries our diathesis-stress dynamics is adopted from Hesp et al. (2021) (with its evidence changed from free energy to valence, §7). The contribution is thus an integration, a task-specific model, and an inferred temporal frame that none of the three has, not a claim to have out-built any one of them on its own terrain; §9 shows the channel decomposition precisely.

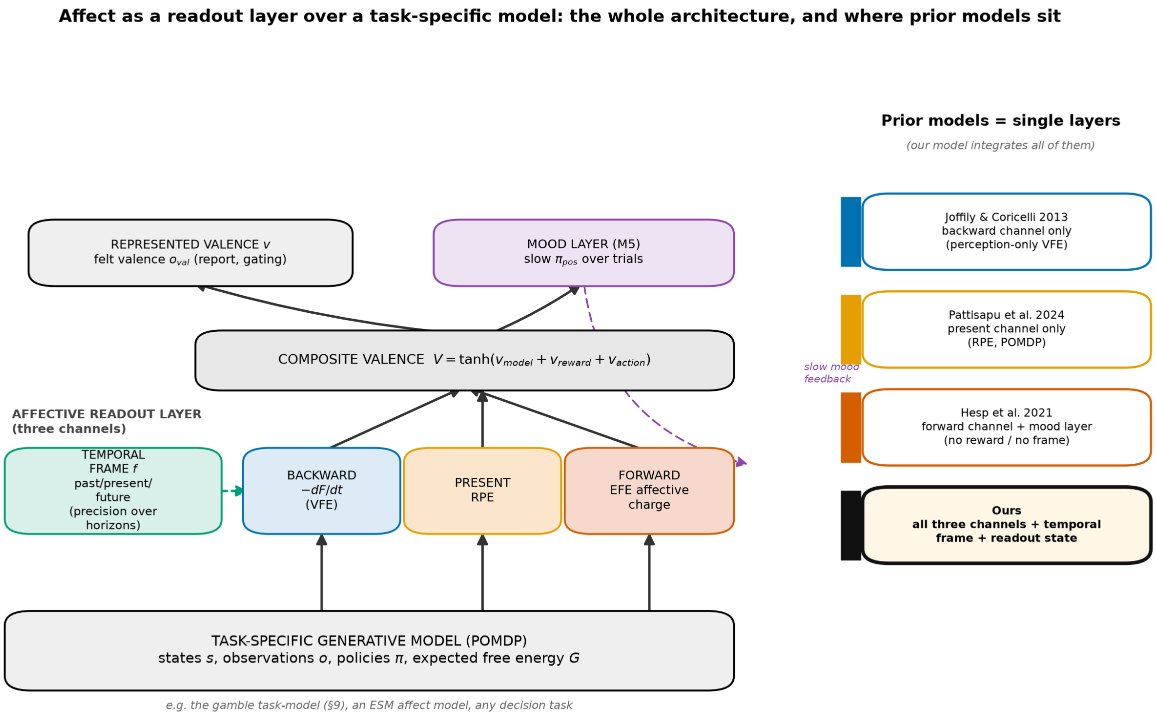

Figure 2: **The layered architecture, and where prior models sit.** Affect is a readout layer over a task-specific generative model (base). The three channels (backward, present, forward) read off the dynamics of inference; their composite feeds a represented valence state and a slow mood layer, with the temporal-frame factor modulating precision across horizons. Each prior model corresponds to a single layer: Joffily & Coricelli to the backward channel, Pattisapu et al. to the present channel, Hesp et al. to the forward channel plus mood layer. Our model integrates all of them and adds the temporal-frame factor and the readout state.

7 Temporal framing dynamics: self-regulation vs self-amplification

Adaptive flexibility is the capacity to move between framing modes as the situation demands. The loop admits two regimes. Under *self-regulation* it is homeostatic. High VFE triggers corrective action, and positive v_{action} sharpens precision on the corrective trajectory. Under *self-amplification* it is pathological. Framing locks into a single mode; precision on that mode's errors intensifies, and the agent cannot update away. We illustrate the dynamics across three parameter phenotypes used in Figure 4: a *healthy* agent (high π_{pos}, intact interoception), a *depressive* one (low π_{pos}, blunted reward sensitivity), and a *manic* one (low interoceptive precision and disinhibited forward framing).

7.1 RECALL supports engagement, and its feedback reliance

Past-oriented framing is preparatory: RECALL stabilises beliefs by pulling toward the prior, lowering VFE and making ENGAGE more competitive, so a RECALL-to-ENGAGE pipeline emerges in healthy agents (Figure 4d,e). This capacity is constrained by feedback. When π_{pos} is low, RECALL fails at two coupled levels: the mechanism itself is impaired ($\alpha \approx 0.14$, barely pulling toward the prior) and its deployment is suppressed (it becomes EFE-uncompetitive).

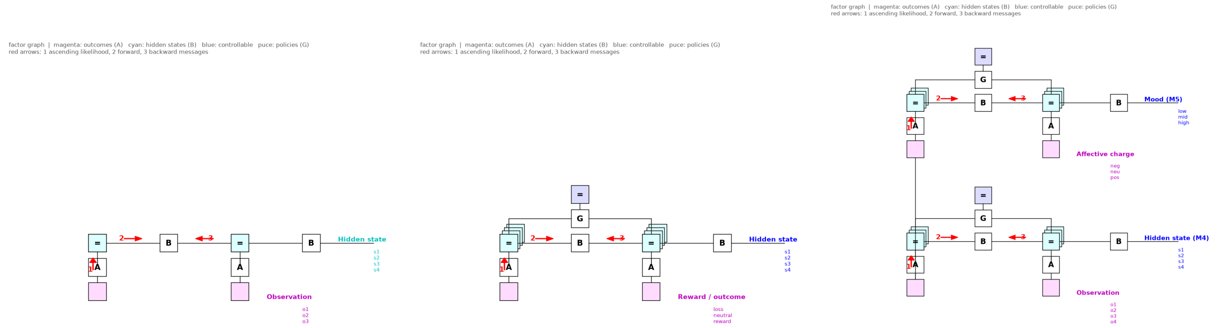

Figure 3: **Predecessor models as reduced factor graphs.** Joffily & Coricelli (backward, perception-only); Pattisapu et al. (present, single-factor POMDP); Hesp et al. (forward, deep hierarchy). Each captures one channel; none has the temporal-frame factor of Figure 1.

Figure 5 contrasts a healthy and a recall-impaired agent differing only in π_{pos}: RECALL utilisation collapses from 26% to $< 1\%$, with compensatory FEEL, ENGAGE, and ABSTRACT. This is the computational form of overgeneral memory (Williams et al., 2007): cognition migrates to representations that are abstract without being self-congruent.

7.2 Chronic stress as maladaptive temporal fixation

Chronic stress realises the double bind of §2.4. Affect turns negative but construal stays coarse, so neither pathway to policy selection delivers. A stressed profile (very high volatility, amplified reward sensitivity, stress-impaired interoception, reduced π_{pos}) shows future-dominated fixation, dominant FUTURE frame belief (0.57 vs 0.32 healthy), depleted PRESENT (0.32 vs 0.46), forward-framing dominance (ABSTRACT, the low-cost prospection operator, taken on 66% of steps), and negative reward valence (-0.35 vs $+0.06$ healthy) from *frustrated forward projection*: abstract prospection commits to above-neutral predictions that a volatile world falsifies (Figure 6). The direction of these effects holds across seeds.

At the slow mood timescale, the M5 layer accumulates believed-valence evidence, so persistently below-neutral affect drives positive-belief precision π_{pos} below the rumination knee, where RECALL turns to rumination and sustains the low-mood state. This produces a *diathesis-stress* pattern rather than a stress-alone one (Figure 7). Crossing vulnerability (low π_{pos}, blunted reward sensitivity, weak interoception) against chronic stress (high volatility) in a 2×2 design, only the conjunction collapses. Across eight seeds a healthy agent stays high even under chronic stress (final $\pi_{\text{pos}} = 5.2 \pm 0.7$, depressed in 0/8), vulnerability alone recovers (4.2 ± 1.0, 0/8), and only the vulnerable agent under stress falls into a self-sustaining depressive attractor (0.8 ± 0.2, 8/8 below the knee). Neither factor alone suffices, matching the diathesis-stress account of depression (Monroe and Simons, 1991) and the resilience of most people to stress. We present this as a mechanistic demonstration, not evidence: the parameters instantiate vulnerability rather than fit it. It makes a falsifiable prediction: in longitudinal experience-sampling, the interaction of vulnerability and stress, not either main effect, should predict transition into sustained low mood, and the transition should be preceded by a persistent below-neutral shift in momentary valence rather than in prediction error.

8 Counterfactual emotions (a generative mechanism)

Regret and relief require comparing what happened with what *would* have happened. Adapting the counterfactual free-energy machinery of Hesp et al. (2020), we write Regret $= F_{\text{actual}} - \min_{\pi'} F_{\text{counterfactual}}(\pi')$. Computing $F_{\text{counterfactual}}$ requires running the generative model offline, exactly what temporal framing does. This mechanism makes a behavioural prediction that a

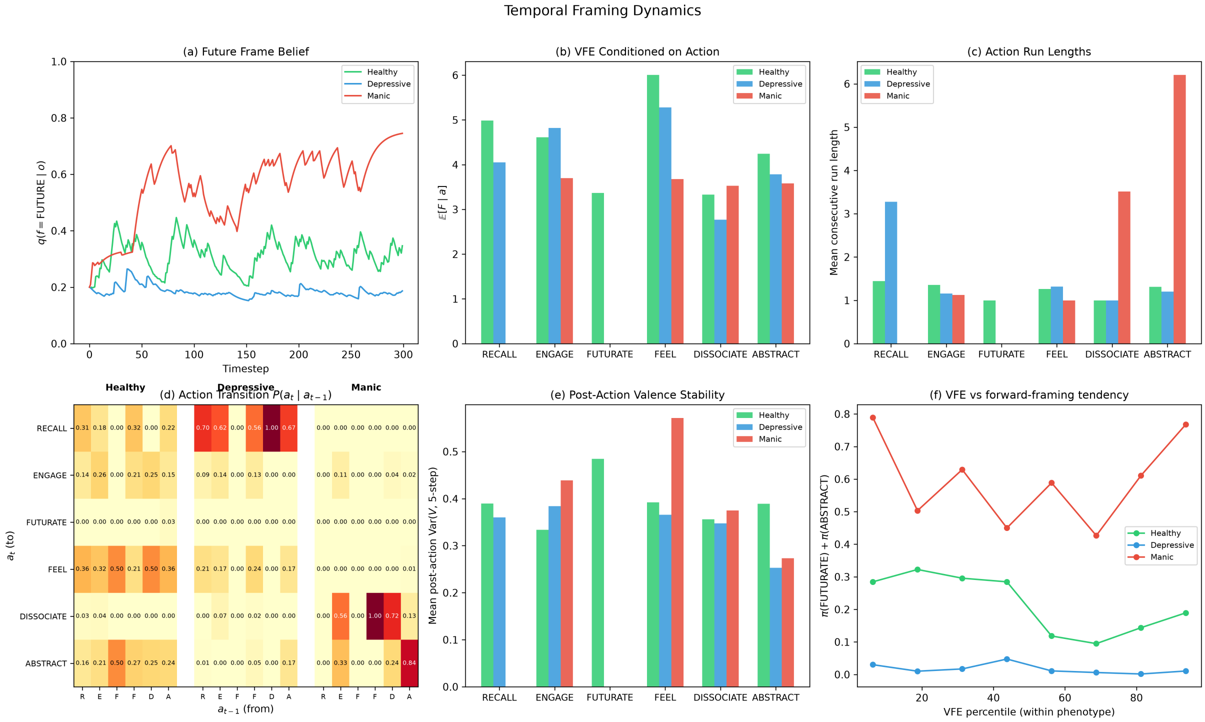

Figure 4: **Temporal framing dynamics.** (a) Future-frame belief over time per profile. (b) Mean VFE conditioned on the selected action. (c) Action run lengths (ABSTRACT and DISSOCIATE longest in the manic profile). (d) Action transition matrices: the RECALL-to-ENGAGE pipeline in healthy agents. (e) Post-action valence stability: RECALL lowest. (f) Forward-framing tendency ($\pi(\text{FUTURATE}) + \pi(\text{ABSTRACT})$, the two future-pulling operators, of which ABSTRACT carries most of the mass) against within-profile VFE percentile. FUTURATE itself is a rare, high-cost reactive-planning burst; ABSTRACT is the low-cost operator that does most forward framing.

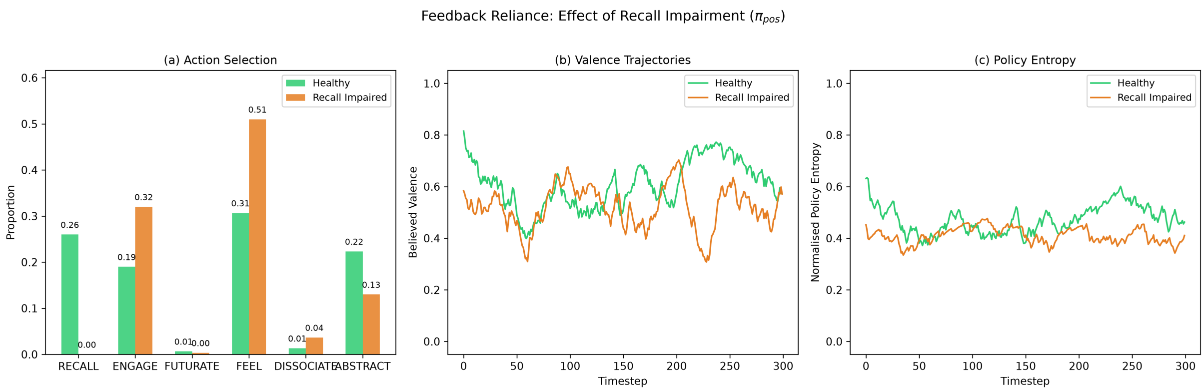

Figure 5: **Feedback reliance under recall impairment.** Profiles share all parameters except π_{pos} (5.0 vs 0.2). (a) RECALL drops from 26% to $<$ 1%, with compensatory FEEL/ENGAGE/ABSTRACT. (b) Valence trajectories lower under impairment. (c) Policy entropy: the impaired agent is less decisive.

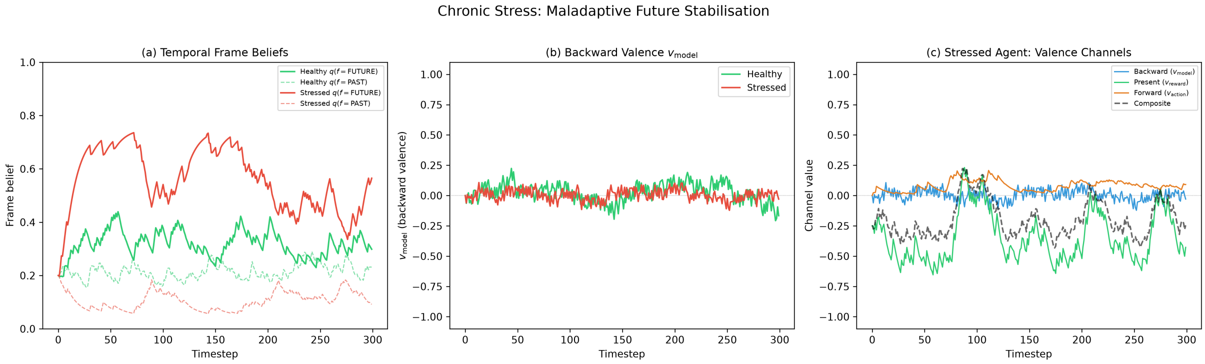

Figure 6: **Chronic stress as maladaptive temporal fixation.** Stressed vs healthy. (a) Frame beliefs: dominant FUTURE, depleted PRESENT. (b) Backward valence more variable under high volatility. (c) Channel decomposition: negative reward valence drives composite valence down via frustrated forward projection.

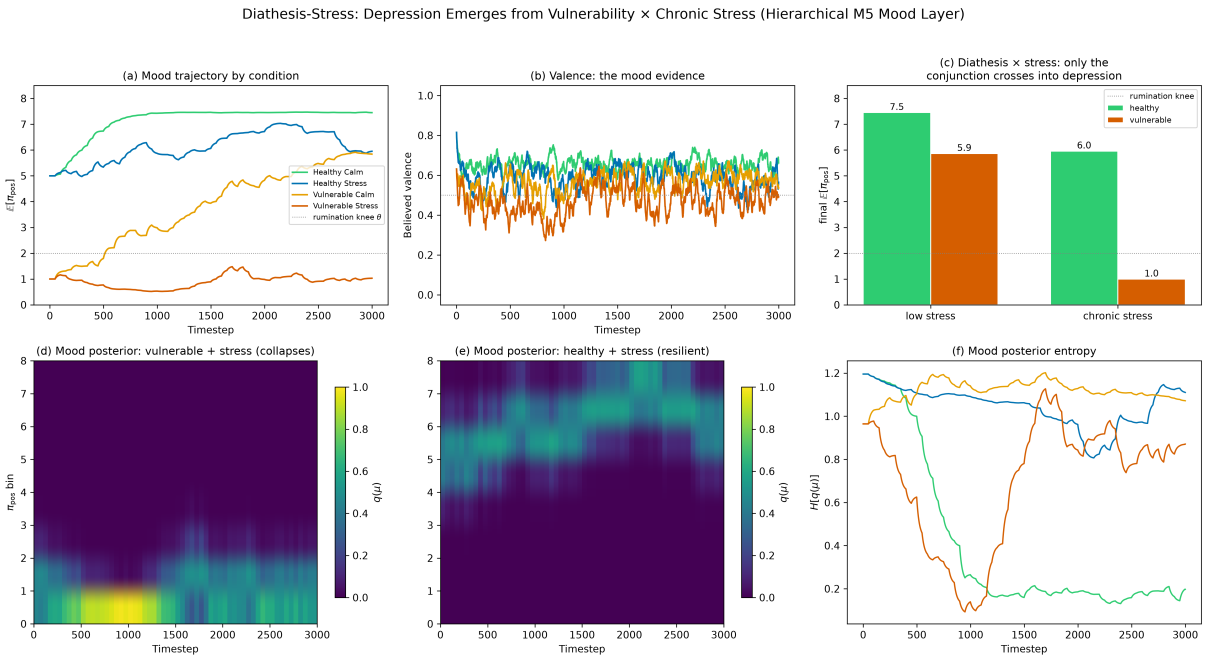

Figure 7: **Diathesis-stress in the M5 mood layer.** A 2×2 design crosses vulnerability (low π_{pos}, blunted reward sensitivity, weak interoception) with chronic stress (high volatility). (a) Mood ($\mathbb{E}[\pi_{\text{pos}}]$) trajectories; (b) believed valence, the mood evidence, relative to neutral (dotted). (c) Final mood by condition: only vulnerability $\times$ stress crosses the rumination knee ($\theta = 2$). (d) The vulnerable-under-stress mood posterior collapses to the low bin; (e) the healthy-under-stress posterior stays high (resilience). (f) Mood-posterior entropy. Robust across seeds (vulnerable+stress depressed 8/8; others 0/8). $T = 3000$.

reward-only model cannot. Across 143 participants, a better *foregone* outcome drives switching on the next choice over and above one's own outcome ($t = 10.4$; $P(\text{switch} \mid \text{regret}) = 0.45$ vs $P(\text{switch} \mid \text{relief}) = 0.26$; Figure 8), a dissociation a factual (no-counterfactual) model has no way to produce. The same switching signature replicates on a second complete-feedback dataset (Palminteri et al., 2017) ($n = 20$). What the counterfactual term does *not* do is improve the prediction of affect magnitude or choice likelihood: on complete-feedback choice data and on large-scale affect data it is redundant with the reward-prediction signal (§9). We therefore class it as a grounded generative mechanism with a specific behavioural signature, not a general predictive-fit improver.

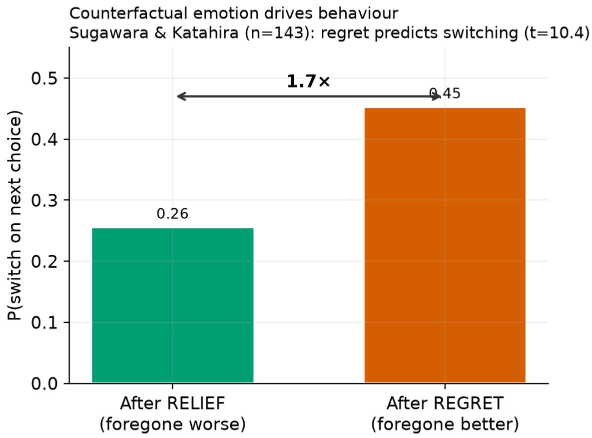

Figure 8: **Counterfactual emotion drives behaviour.** People switch ∼1.7× more after regret than relief (Sugawara and Katahira, 2021, $n = 143$), a signature a factual model cannot generate. This validates the mechanism as the generator of regret/relief, independent of its (null) contribution to prediction.

9 Empirical validation: head-to-head and extension

We tested the model on open data, holding out participants throughout. Provenance, scripts, and full results are in the supplementary `EMPIRICAL_RECORD.md`; Table 2 lists the datasets and what each supports.

9.1 Subsumption: our affect layer recovers the happiness equation

We instantiate our three channels over a *gamble* task-model (policies {safe, gamble}; preferences increasing in reward) for the Great Brain Experiment happiness data (Rutledge et al., 2014) (47,067 participants; evaluated held out across a 14,803-participant subset, 96,624 held-out ratings). We reconstruct each predecessor as its single affective channel over this shared task-model (single-channel reconstructions, not the authors' original implementations) and compare them against the integrated readout on the same held-out ratings (the task-model is Figure 9, its combination with the emotion model is Figure 10; results in Table 3 and Figure 11A). No single channel reproduces the integrated result. The present channel (Pattisapu, RPE) reaches $R^2 = 0.111$ and the forward channel (Hesp, EFE) $R^2 = 0.034$, while the backward channel (Joffily, $-dF/dt$) scores $R^2 = 0.000$. The last is expected rather than a defeat: a rate-of-change signal cannot predict a happiness *level* on a single-shot gamble that has no temporal dynamics to track. The integrated model reaches $R^2 = 0.144$, recovering the standard happiness equation (CR+EV+RPE, $R^2 = 0.146$). No expected-value term is entered as a regressor; expected value instead emerges from the forward/EFE channel, which on this task encodes anticipated reward

Table 2: **Datasets.** N is participants (held-out analysis subset in parentheses); ES = effect sizes.

Dataset	N	Signals	Access	Supports
Rutledge GBE happiness (Rutledge et al., 2014)	47,067 (14,803)	gamble choices; momentary happiness	CC0 (Dryad)	subsumption / affect readout
Geschwind–Bringmann ESM (Bringmann et al., 2013; Geschwind et al., 2011)	130 (129)	momentary affect; event pleasantness; worry	CC BY-NC	affect-dynamics; frame-to-worry
Reliability ESM (`osf.io/83cfk`)	91	momentary affect (12 emotion sliders)	open (OSF)	affect-dynamics replication
Temporal-orientation ESM (Mulholland et al., 2023)	101	per-probe past/future thought; valence	CC BY-NC	direct frame-to-affect coupling
Sugawara & Katahira (Sugawara and Katahira, 2021)	143	choices; obtained + foregone outcomes	open (OSF)	counterfactual signature
Palminteri et al. (Palminteri et al., 2017)	20	choices; obtained + foregone outcomes	open (OSF)	counterfactual replication
Reward+punishment reversal (osf.io/2gq96)	128 (64/64)	learning rates by valence	open (OSF)	general precision deficit
Reward Positivity (MEG) (Pirrung et al., 2025)	90 (52/38)	neural reward response	CC0	reward reactivity (c_{pos})

by construction, and adding it to the present channel raises R^2 from 0.111 to 0.144 (+30%). The happiness equation is thus a special case of our framework.

9.2 Extension: the full model predicts affect dynamics, on two samples

Where the task has temporal structure, the temporal-frame factor earns its keep, and we report the advantage on two independent experience-sampling samples rather than one. In a remitted-depression ESM sample (Bringmann et al., 2013; Geschwind et al., 2011) (129 participants held out; global parameters fitted on training participants and evaluated on unseen ones), the full model predicts next-moment valence with held-out $R^2 = 0.19$ at one step versus 0.09 for the best linear/autoregressive and event-regression baselines, roughly double the explained variance at this horizon (Figure 11B), with smaller leads at two and three steps. In a second, independent reliability ESM sample (`osf.io/83cfk`, 91 participants, no event or worry items, so the model runs on valence alone), the model again leads at every horizon, but the margins are modest and the one-step gap is near parity (Figure 11C, Table 4). The two samples differ in exactly the way the framework predicts matters: in the reliability sample affect is already highly persistent (baseline one-step $R^2 = 0.475$ versus 0.089), leaving little non-trivial dynamics for temporal machinery to capture, whereas the clinical sample's more volatile affect is where the model helps most.

Component ablations locate the source of the advantage. Driving the model on valence alone in the clinical sample changes almost nothing ($R^2 = 0.20$ at one step), so the gain is not the event channel acting as a hidden regressor. Removing the persistence-like valence-inertia term leaves the clinical-sample advantage intact (the model still reaches $R^2 = 0.14$ at one step and 0.29 at two steps with no inertia at all, versus baselines 0.09 and 0.27); on the already-persistent reliability sample the model sits at parity without inertia (0.46 vs 0.48 at one step, 0.34 vs 0.34

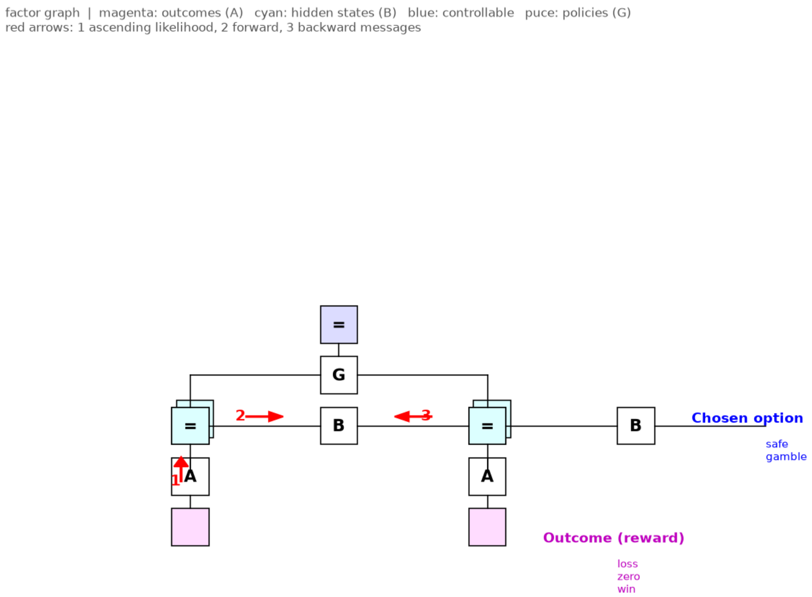

Figure 9: **The gamble task-model used for the subsumption test.** The task-specific generative model over which the affect channels are read out: a single reward-outcome modality (loss/zero/win) generated by the chosen option (safe/gamble), with a preference gradient increasing in reward. Affect is a readout layer over this model, and each predecessor corresponds to one channel of that readout. This is the gambling-task instance of the general architecture in Figure 1; Figure 10 shows the two joined.

Table 3: **Channel decomposition on the Rutledge gamble.** Each predecessor is instantiated as its single affective channel over the same gamble task-model (Figure 9); held-out R^2 for momentary happiness (14,803 participants, 96,624 ratings), no expected value injected as a regressor. On a single-shot gamble the backward (rate-of-change) and forward (policy-revision) channels have no temporal dynamics to track, so their low scores are category-appropriate, not defeats; the informative comparison is the present channel alone (0.111) against the integrated readout (0.144), which reaches the reference ceiling.

Model	Channel	Held-out R^2
Joffily (Joffily and Coricelli, 2013)	backward (VFE rate)	0.000
Hesp (Hesp et al., 2021)	forward (EFE charge)	0.034
Pattisapu (Pattisapu et al., 2025)	present (RPE)	0.111
Ours	**all three**	**0.144**
Happiness equation (Rutledge et al., 2014)	CR+EV+RPE (reference)	0.146

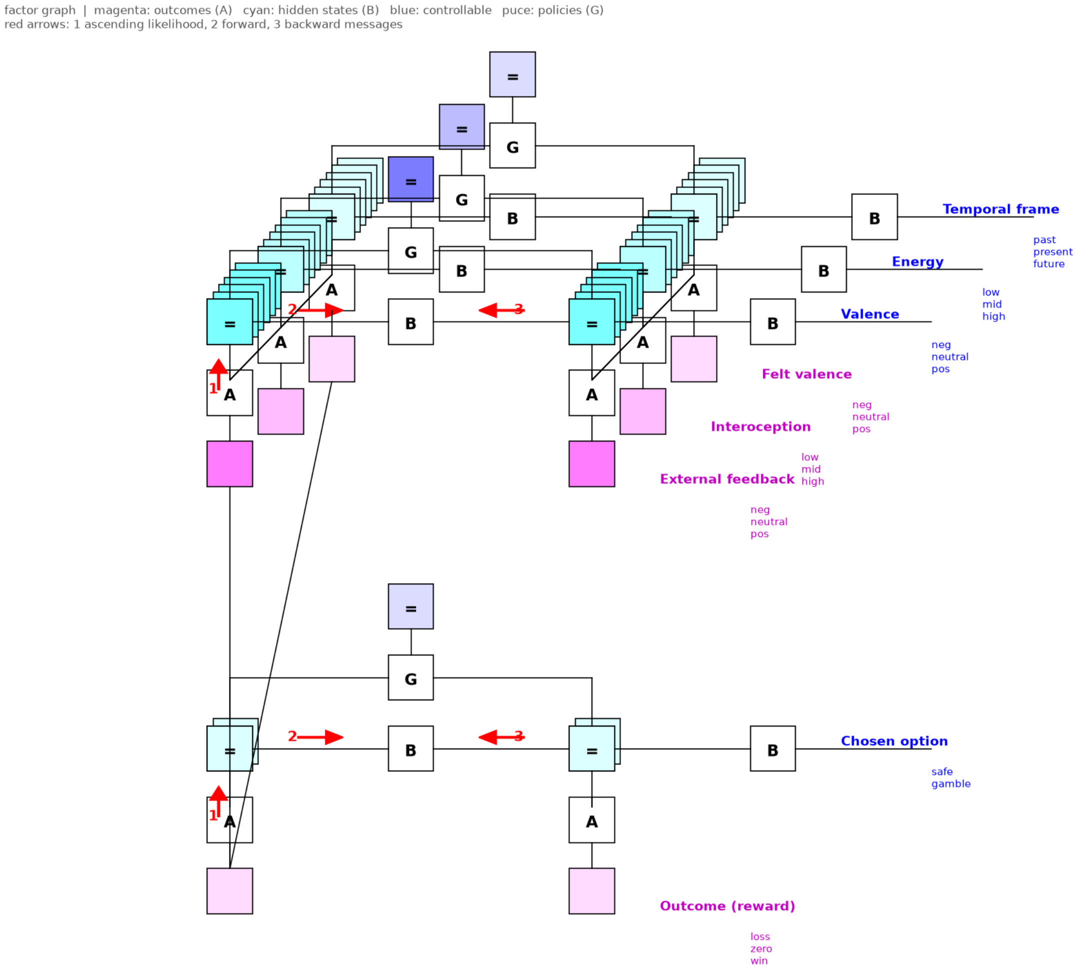

Figure 10: **The gamble task-model joined with the emotion model.** Two-level factor graph of the subsumption test as actually run. The gamble task-model of Figure 9 (lower level: chosen option generating the reward outcome) sits beneath the emotion model of Figure 1 (upper level: valence, energy, and temporal frame with their three modalities). The connecting edges carry the task's reward outcome into the affect layer, directly as external feedback, and via the three channels reading the task-model's inference dynamics as the felt-valence observation. Each predecessor model in Table 3 corresponds to retaining a single channel of this upper layer.

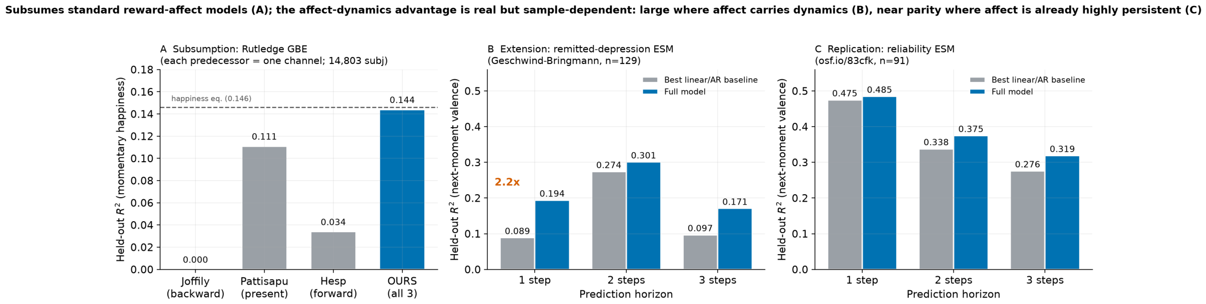

Figure 11: **A general affect-readout framework.** (A) Channel decomposition (Rutledge GBE): each predecessor reconstructed as one channel over the same gamble task-model; the present channel (Pattisapu 0.111) is the informative single-channel comparison, the backward and forward channels being category-mismatched on a static gamble, while integration reaches the reference ceiling (0.144 vs happiness equation 0.146, dashed). (B) On the remitted-depression ESM sample, the full model reaches 2.2× the held-out R^2 of the best linear/autoregressive baseline at one step. (C) On an independent reliability ESM sample the model leads at every horizon but by smaller margins, near parity at one step, because affect there is already highly persistent (baseline $R^2 = 0.475$). All held out across participants; Table 4.

at two), consistent with there being little non-persistence structure to capture there. Where the advantage exists, the model's own forward dynamics rather than a persistence prior carry it. This holds because the precision-gated prospection operator (§4) keeps forward rollouts calibrated to the agent's actual affective level rather than drifting toward a fixed optimistic target. We do not claim a head-to-head reimplementation of each predecessor here; the claim is that the integrated model outpredicts standard baselines where affect carries non-trivial dynamics, and single components are insufficient.

Table 4: **Affect-dynamics prediction on two independent ESM samples.** Held-out R^2 for next-moment valence by prediction horizon, full model versus the best linear/autoregressive baseline on identical inputs, 5-fold cross-validation with whole participants held out. The reliability sample has no event items, so the model runs on valence alone there.

Sample	Predictor	1 step	2 steps	3 steps
Remitted-depression (n=129)	baseline	0.089	0.274	0.097
(Bringmann et al., 2013; Geschwind et al., 2011)	**full model**	**0.194**	**0.301**	**0.171**
Reliability (n=91)	baseline	0.475	0.338	0.276
(`osf.io/83cfk`)	**full model**	**0.485**	**0.375**	**0.319**

9.3 Non-circular latent-state validation

Driven only by valence and event inputs, the model's posterior *future-frame* belief covaries with an independently measured worry item it is never shown (r = 0.166, n = 11,712). We read this as evidence that the frame belief couples to a symptom-relevant affective signal, not that it recovers reported temporal orientation as such; the direct orientation test below makes that distinction concrete.

9.4 Direct temporal-orientation test

The strongest test of the temporal-frame construct uses a dataset that labels the orientation of thought itself. In an experience-sampling study with per-probe reports of past- and future-directed thought alongside momentary valence (Mulholland et al., 2023; $n = 101$, 1,458 probes, CC BY-NC), the model's central structural prediction holds and one of its default assumptions fails. The prediction that past-framing carries negative valence under low positive-belief precision (the RECALL/rumination branch) holds directly and within person: reported past-orientation covaries negatively with momentary valence ($r = -0.23$ within person), and the effect attenuates for people with higher trait positivity ($\beta_{\text{past}\times\text{trait}} = +0.06$), the direction the precision-gated coupling predicts. The default that *future*-framing is optimistic fails: future-orientation is mildly negative in this unselected sample ($r = -0.07$). This motivated the precision-gated FUTURATE operator of §4, in which low precision yields worry-type prospection. Two limits remain. First, the gating softens the negativity of past-framing without reversing it, so the strong reading in which rumination flips into positive narrative stabilisation overstates a real but weaker effect. Second, driven on affect alone the model's latent frame does not *recover* reported orientation ($r \approx 0$), because orientation is exogenous context the model is never given; the validated claim is a frame-to-affect coupling, and testing affect-to-frame recovery would require an orientation-carrying input channel.

9.5 What does not improve prediction

Across every dataset we tested, experience-sampling affect, two complete-feedback counterfactual-learning tasks, a probabilistic reward task, and the large-scale happiness data, and for both choice and affect targets, neither the hedonic asymmetry ($c_{\text{pos}} \neq c_{\text{neg}}$) nor counterfactual depth improves prediction. Both are redundant with reward-prediction signals. We therefore present them as *generative* and *neurally/experientially* grounded mechanisms rather than predictive components (Table 5).

Table 5: **What the framework can and cannot claim.** Predictive claims are out-of-sample or out-of-model gains; generative claims are mechanisms that reproduce phenomena and are neurally or experientially grounded but do not improve behavioural prediction.

Claim	Status	Evidence
Multi-channel integration	predictive	recovers the happiness equation (Table 3); outpredicts baselines on two ESM samples, $\sim 2\times$ at one step where affect is volatile (Table 4)
Latent temporal-frame state	predictive	future-frame belief tracks an unshown worry item ($r = 0.17$)
Counterfactual emotion	generative	regret drives choice-switching ($t = 10.4$); adds no choice or affect prediction
Hedonic asymmetry ($c_{\text{pos}} \neq c_{\text{neg}}$)	generative	blunted Reward Positivity, self-reported anhedonia; not recoverable from choice

10 Clinical mapping: a two-level account of depressive affect

The framework draws a two-level picture of depressive affect and, more usefully, predicts a dissociation between the levels that the data bear out. (i) A *general* reduction in reinforcement-learning rate/precision, present for both reward and punishment learning. In our analysis of an open reward-and-punishment reversal-learning dataset (`osf.io/2gq96`), depressed participants showed reduced learning rates in *both* conditions (Cohen's $d \approx -0.5$ to -0.6, not reward-selective), which maps onto reduced overall precision and replicates the established mapping of anhedonia onto reinforcement-learning parameters (Huys et al., 2013). (ii) A *reward-specific* valuation/reactivity deficit ($c_{\text{pos}} \downarrow$), which converges with reward-specific ventromedial-frontal hypoactivation and blunted Reward Positivity (Pirrung et al., 2025) and with self-reported anhedonia. The two levels dissociate exactly as the model requires: c_{pos} scales the *value* an outcome is assigned, while the learning rate governs how fast beliefs update, so a sample can show reward-specific reactivity blunting (level ii) alongside a general learning-rate deficit (level i) without contradiction, and the Pirrung sample does. The value deficit is what Reward Positivity indexes; the learning deficit is what the reversal task indexes. We label (ii) a model-derived prediction with convergent neural and self-report support that awaits direct behavioural confirmation. A companion paper develops the full bipolar phenotype characterisation.

11 Discussion

11.1 What the integration buys

The single move behind every result here is treating affect as a readout over a task-specific model rather than as a model in its own right. That move has an empirical consequence and a conceptual one. Empirically, the same three channels that recover the happiness equation on a static gamble gain predictive power once the task carries temporal structure, because only then does the temporal-frame factor have work to do. Conceptually, prior single-channel accounts stop being rivals and become limiting cases: each is what our readout reduces to when its task offers no dimension for the other channels to read. Where a task lacks temporal structure the framework reduces to the task-appropriate reward-affect model.

11.2 Metastability as health

Health is not a particular temporal configuration but the *capacity to transition* between configurations. The healthy agent visits multiple framing modes with a high switching rate and short run lengths, metastability, in dynamical-systems terms. Pathology is stickiness. Amplified reward sensitivity with impaired interoception locks abstract forward framing on via absorptive narrative inertia; a habit prior locks RECALL on while suppressing EFE-optimal escape. Restoring the ability to transition matters more than correcting the balance between orientations.

11.3 Attention as precision, and mood/emotion timescales

Temporal framing extends attention-as-precision (Feldman and Friston, 2010) to the temporal domain: $\mathbf{B}_{\text{frame}}$ matrices are temporal-attention operators, and framing is deciding *when* to attend. The framework sits at the interface of the mood/emotion timescale hierarchy (Hesp et al., 2021). Individual frame selections are within-trial (M4) phenomena; sustained biases are cross-trial (M5) mood. This is where the thin (affect signals free-energy trajectories) and thick (affect is the enactive force directing attention across horizons) readings of emotion meet.

11.4 Limitations

Emotion-level policy selection is single-step (Friston et al., 2018); the frame is categorical, not yet a continuous depth; the E-vector is fixed and not yet learned (Watkins, 2008); and the model is single-agent, with dyadic temporal co-regulation (Hinrichs et al., 2025) left to future work. Empirically, the per-channel head-to-head is on one dataset (Rutledge) and awaits replication; the affect-dynamics advantage replicates across two ESM samples but with sample-dependent margins. The direct temporal-orientation test (§9.4) validates the frame-to-affect coupling but also shows its two current limits: the model does not invert reported orientation from affect alone (it is not given orientation-carrying input), and precision gating attenuates rather than reverses the negativity of past-framing. Grounding the frame in an orientation-carrying observation channel, so the latent frame can be tested for recovery and not only coupling, is the most valuable next step.

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